

Problem 5.6

The sum of \$5,000 is invested for the months of April, May, and June at 7% simple interest. Find the amount of interest earned:

- (a) assuming exact simple interest in a non-leap year,
- (b) assuming exact simple interest in a leap year (with 366 days),
- (c) assuming ordinary simple interest,
- (d) assuming Banker's Rule.

We use the simple interest formula:

$$I = Prt,$$

where $P = 5000$, $r = 0.07$, and t is the time in years.

Number of Days

Using the counting shown in the exercise, the exact time for April, May, and June is:

$$29 + 31 + 30 = 90 \text{ days.}$$

For ordinary simple interest, each month is taken as 30 days, so:

$$30 + 30 + 30 = 90 \text{ days.}$$

(a) Exact Simple Interest in a Non-Leap Year

Using 90 days and a 365-day year:

$$I = 5000(0.07) \left(\frac{90}{365} \right).$$

Therefore,

$$I \approx 86.30.$$

(b) Exact Simple Interest in a Leap Year

Using 90 days and a 366-day year:

$$I = 5000(0.07) \left(\frac{90}{366} \right).$$

Thus,

$$I \approx 86.07.$$

(c) Ordinary Simple Interest

Using 90 days and a 360-day year:

$$I = 5000(0.07) \left(\frac{90}{360} \right).$$

So,

$$I = 87.50.$$

(d) Banker's Rule

Using the exact 90 days and a 360-day year:

$$I = 5000(0.07) \left(\frac{90}{360} \right).$$

Hence,

$$I = 87.50.$$